

**Spatial Exploration and Descriptive Analysis of Survey, Crime, and Census Data for
Mental Health Research**

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Executive Summary

This project explored how county-level social and economic conditions relate to mental-health outcomes across the United States. By integrating demographic, income, crime, and survey data, this project produced two complementary visualization tools.

Data from the FBI Crime Data Explorer, the American Community Survey, and national mental-health survey responses were combined by county identifiers to create a unified dataset. ArcGIS Pro was used to clean, standardize, and integrate the geographic and demographic data across sources, ensuring alignment at the county level. The cleaned dataset was then imported into R Studio, where additional variables were created to support analysis and visualization in the R Shiny application. These included simplified race and education categories, as well as percentage-based indicators for depression, anxiety, self-harm, suicidal attempts and suicidal ideations, allowing for consistent comparison across counties. The final dataset served as the foundation for both the ArcGIS spatial mapping and the R Shiny interactive dashboard.

The ArcGIS maps were used to contextualize survey respondents within their broader geographic and social environments rather than to analyze mental health outcomes directly. By overlaying respondent locations with county-level crime, demographic, and income data, the maps illustrate how participants are distributed across differing social and structural contexts. Urban and densely populated counties tended to have higher crime rates and greater racial diversity, while counties in the Midwest and Great Plains were characterized by lower crime levels and more homogenous populations. Median household income also exhibited clear regional patterns, with higher values concentrated along the Northeast and West Coast and lower values across much of the rural South. Survey responses were most frequently located in urban

areas, highlighting uneven geographic coverage and helping identify the types of county-level environments represented in the survey data.

The R Shiny dashboard expanded on these findings by highlighting demographic differences in mental-health outcomes. Males reported higher rates of suicide attempts and self-harm, whereas females reported more suicidal ideation. Younger adults, particularly those aged 18 to 29, exhibited elevated anxiety and depression. Respondents identifying as homosexual and bisexual experienced substantially higher rates of suicidal ideation and self-harm than heterosexual respondents. Distinct risk profiles also emerged by employment, education, and insurance status, suggesting that economic and health-care access factors shape mental-health experiences.

While the survey data reflect a geographically uneven sample and are not intended to be population-representative, the ArcGIS map improves interpretability by integrating respondent locations with county-level demographic, income, and crime data. This approach enables a spatially informed examination of the environments captured by the survey and highlights structural variation across regions.

1. Introduction

Understanding how social and environmental factors shape mental-health outcomes is a central focus in psychiatric epidemiology, particularly among young adults whose experiences of stigma, help-seeking behavior, and overall well-being are influenced by local socioeconomic and demographic contexts (WHO, 2014). Factors such as community composition, crime exposure, and income inequality can interact with individual behaviors to affect mental-health experiences (Sampson et al., 1997).

This project was conducted in collaboration with Dr. Hans Oh and his team, who are in the preliminary stages of analyzing how local environments relate to mental-health outcomes. The team provided survey data collected from Black and Asian/Pacific Islander American emerging adults, covering measures of mental-health symptoms, stigma, help-seeking intentions, substance use, food insecurity, and awareness of mental-health services. To situate these survey responses within a broader social context, external demographic, income, and crime data were incorporated to represent each respondent's local environment.

At the exploratory stage, visualization tools can be highly effective for identifying contextual patterns and guiding subsequent analyses. Therefore, the goal of this project was to create robust, interactive visualization tools that integrated individual level survey data with county level characteristics such as crime rates, racial composition, and median household income to represent respondents' local environment.

The following sections describe how the combined survey and contextual datasets were processed and visualized through two tools: ArcGIS Pro web map and R Shiny dashboard.

2. Methods

The methodological workflow consisted of three main components: data preprocessing, ArcGIS mapping, and R Shiny application development. ArcGIS is a geographic information system (GIS) software platform used to manage, analyze, and visualize spatial data through interactive maps. R Shiny is a web application framework that enables the creation of interactive data dashboards and tables using the R programming language.

2.1 Data Pre-processing

The data preprocessing phase focused on preparing and combining external county-level crime, demographic, socioeconomic data with the client-provided survey data for geographic visualization and analysis. All data cleaning, transformation, and integration procedures were performed in ArcGIS Pro using its point-and-click tools, with only short field-calculation expressions applied when necessary.

The FBI crime dataset required extensive cleaning before it could be joined to county boundaries (Federal Bureau of Investigation, 2024). The original Excel workbook contained formatting features that interfered with import, so it was first converted to CSV format. After imported, inconsistencies in county and agency names were corrected to ensure successful joins with the county geography data. Duplicated entries were examined: when one record contained all zeros and another contained valid data, the valid record was retained; when both held differing values, county-level crime reports were revised to identify the most reliable entry. Total crime counts were then computed for each county by summing across all categories, and percentages were calculated to determine each county's "majority crime category", which is the crime type with the highest proportion of total reported offenses.

The ACS demographic data required several standardization steps (U.S. Census Bureau, 2023). The original files contained extensive coded metadata, which were replaced with descriptive variable labels for clarity. All fields stored as text were converted to numeric format to enable quantitative analysis. To focus on the most relevant information, only key variables were retained, including county and state names, geographic identification numbers (GEOID), estimated total populations, estimated populations by race, and estimated male and female populations for each race. After cleaning, race-specific percentages were calculated by dividing each racial population by the county total. These percentages provided a normalized basis for comparing demographic composition across counties. Finally, all race-specific tables were merged into a unified county-level demographic dataset keyed by geographic identifiers and joined to the county shapefile for mapping.

Median household income data by race were processed in the similar way (U.S. Census Bureau, 2023). Column names were standardized using descriptive labels, and unnecessary columns, including blank fields and margin-of-error variables, were removed to improve clarity. Each race-specific income table was merged into a unified county-level income dataset. These income variables were joined to the county geography to visualize racial income disparities and socioeconomic context. It is important to note that the U.S. Census Bureau collects income data using the head of household's race to represent the race of the household, which may influence interpretation of racial income differences.

The survey data was already well structured and required only minimal cleaning. Respondent IP addresses were removed to protect privacy, and other nonessential variables were retained but excluded from map display. Each response included latitude and longitude coordinates, which enabled linkage to county boundaries. A spatial join was performed to assign

each respondent to a county, using a stepwise search radius that expanded in 5-mile increments up to 35 miles to correct coordinate inaccuracies. Once all respondents were successfully matched, county-level crime, demographic, and income datasets were added as separate layers, allowing each location to display both individual responses and contextual county characteristics within ArcGIS.

After all spatial joins were finalized, all cleaned external datasets were consolidated with the survey data into a single integrated file. Counties without any respondents were removed to ensure accurate geographic representation. The resulting dataset contained 2,801 rows and 467 variables that integrated county-level demographic, socioeconomic, and crime data for subsequent analysis and visualization. This file was exported from ArcGIS in CSV format, making it available for client use, and was also imported into R Studio, where additional variables were generated to support interactive visualization in the R Shiny application.

From the merged dataset, 22 key variables reflecting the client's primary areas of interest were selected for inclusion in the R Shiny application. The Race variable was recoded into three categories: Asian, Black, and Two or More Races to simplify the original 31 unique values and improve clarity in visualization. The Education and Health Insurance variables were also standardized and shortened to enhance readability in tables and plots.

Additional variables were derived from the merged dataset to deepen analytical interpretation. New variables included county participant totals and a race-specific income ratio, defined as each race's median household income divided by the county median. Demographic percentages were also created for variables such as sex, age, sexual orientation, employment status, race, education, and health insurance. The demographic percentages were analyzed across three outcomes of interest: depression and anxiety, self-harm, and suicidal ideation or attempts

within 12 months. Each percentage was computed relative to the total respondents within its outcome category. For example, the percentage of female respondents who reported self-harm was computed as the number of females with self-harm divided by the total number of respondents who reported self-harm. These percentages ensured a consistent scale for visualization and prevented misleading interpretations that might occur if raw counts were plotted on the y-axis.

2.2 ArcGIS Map

Separate copies of the county layer were created so that crime, race, and income datasets could be symbolized independently without interfering with one another. A separate survey-response layer was added to visualize respondent concentrations across counties. Each layer was designed with interactive functionality. Selecting a county or respondent cluster triggers a detailed pop-up displaying relevant county and respondent information. After integration, multiple county-level map layers were developed to visualize key contextual indicators:

- Crime Data Layer: Displayed as a graduated color map showing total crime counts normalized by county population. (Figure 1)
- Race Layer: Displayed as a categorical color map representing the majority race in each county. (Figure 2)
- Income Layer: Displayed as a graduated color map illustrating each county's median household income. (Figure 3)
- Survey Respondent Layer: Displayed as a heat map visualizing respondent concentration within counties. (Figure 4)

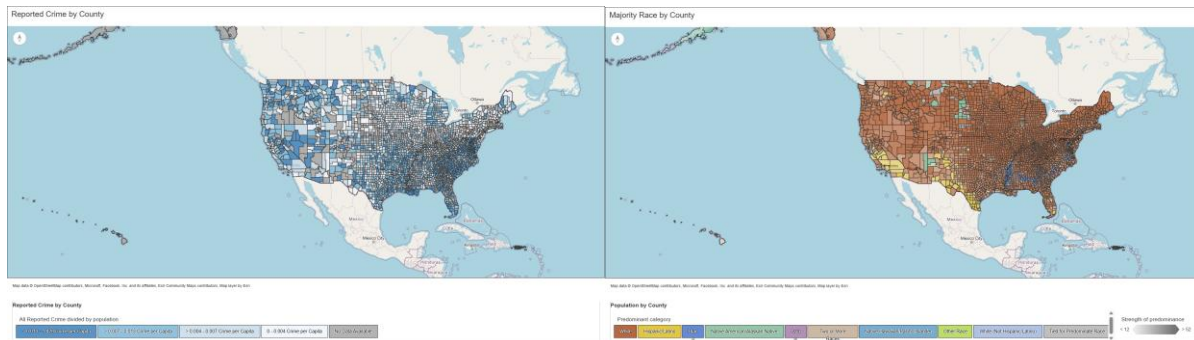


Figure 1: Crime per Capita by County

Figure 2: Most Reported Race & Predominance by County

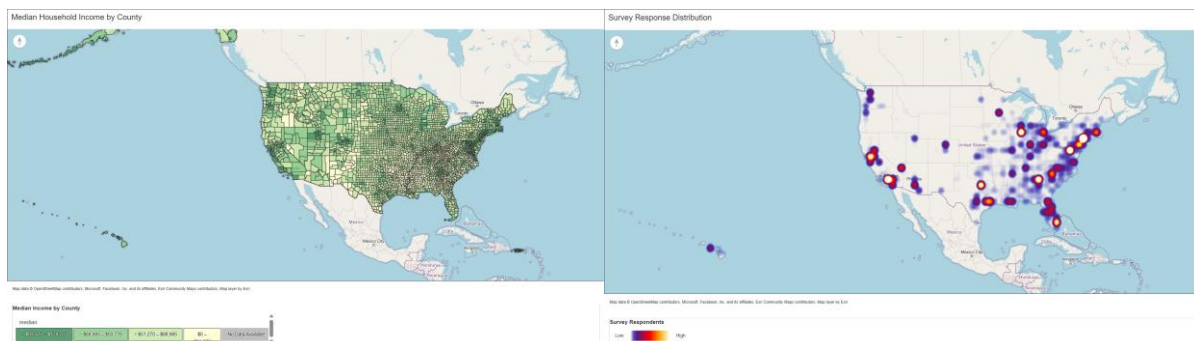


Figure 3: Median Household Income by County

Figure 4: Survey Respondent Distribution

2.3 R Shiny Application

After data integration, an R Shiny application was developed to transform the merged dataset into an interactive exploration tool. The app was built using the *Shiny* and *Shinydashboard* R packages, which integrate the user interface and server logic within a unified environment. Additional packages extended functionality of the application: the *plotly* package created interactive visualizations, the *reactable* package supported dynamic and filterable data tables, the *scales* package formatted numeric values for readability, and the *purrr* package automated repetitive calculations across multiple variables.

When users open the R Shiny application, they are presented with a dashboard-style layout that includes a navigation menu and multiple tabbed views. Each tab displays a different set of tables or visualizations designed to summarize county-level characteristics or compare

mental-health outcomes across demographic groups. User selections, such as choosing a county or selecting a demographic and outcome variable, automatically update the displayed results, allowing patterns to be explored interactively without manual reloading.

The application supports dynamic data exploration through interactive tables and bar charts. Tables allow users to search for specific counties, sort values, and apply filters to focus on variables of interest. Visual summaries are presented as bar charts that display percentage distributions and reveal exact values when hovered over. All visual outputs are scaled consistently from 0 to 100 percent to support clear comparison across groups and avoid misinterpretation due to unequal sample sizes. Demographic percentages displayed in the application were calculated systematically to ensure that all summaries reflect comparable proportions across mental-health outcome categories.

3. Results

3.1 ArcGIS Map

When the user enters the map viewer they are presented with a navigation tip window, explaining how to interact with the map. (Figure 5) After closing the window, the user sees the default display of survey response distribution. The multiple interactive layers in the map that can be toggled on or off, allowing users to explore geographic variation from different perspectives.

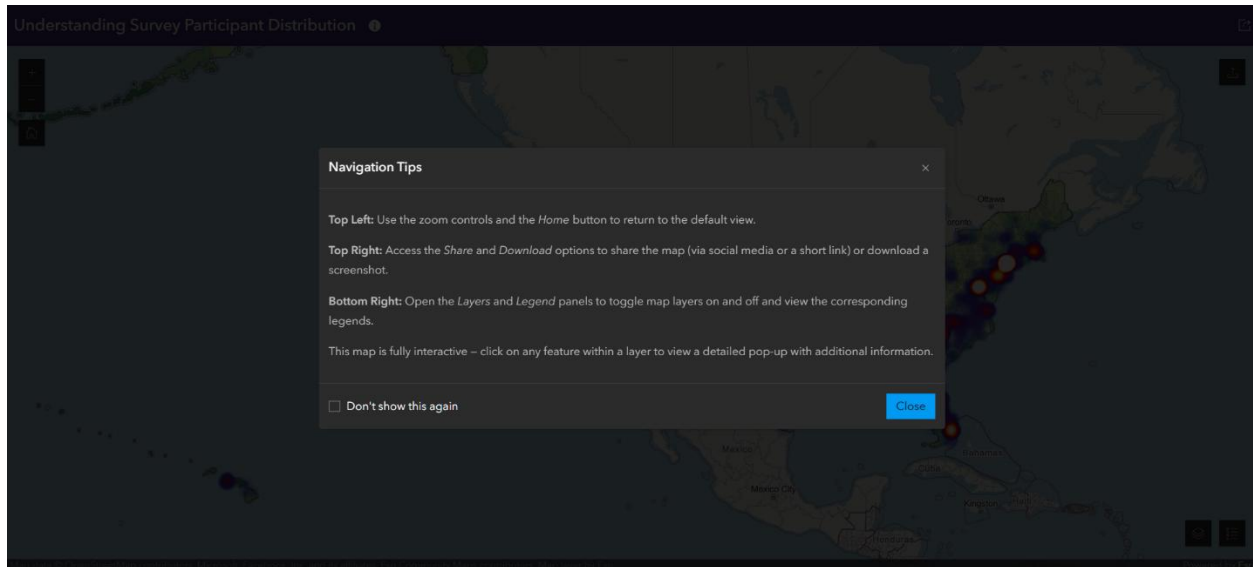


Figure 5: Interactive ArcGIS Map Landing Page

Interactive tools allow users to zoom in and out, reset the map to its default view, share the map through a link, or download the current display. A legend panel identifies map symbols, and the layer toggle enables quick comparisons among different variables. These features make the interface intuitive and adaptable for users with varying levels of experience. (Figure 6)

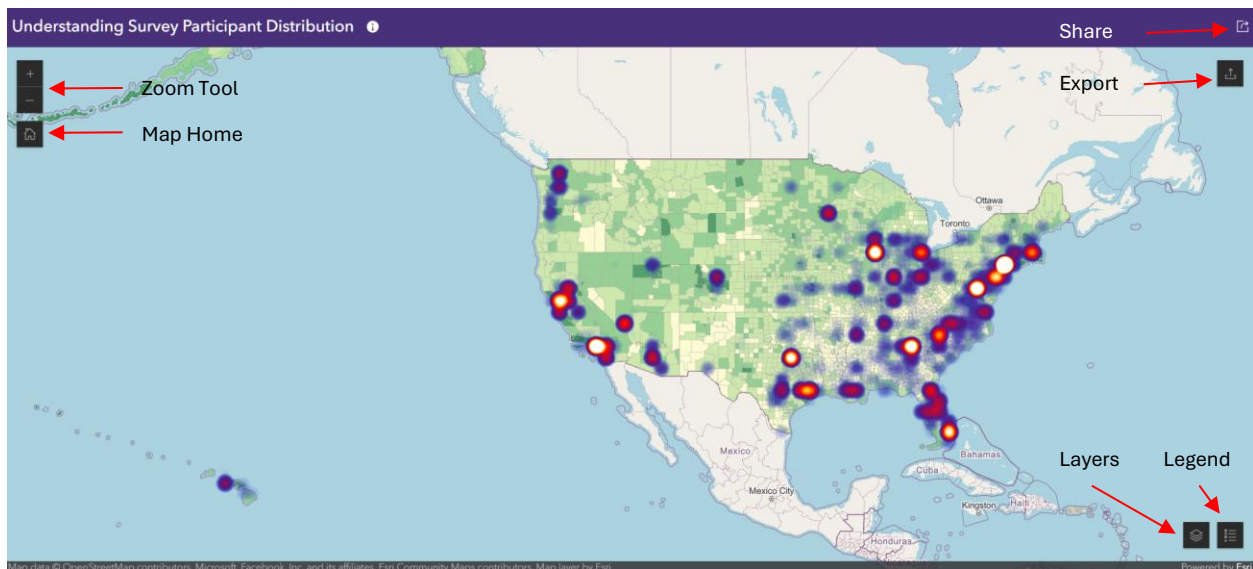


Figure 6: Interactive ArcGIS Map

Each layer is fully interactive. Selecting a county or a respondent cluster opens a pop-up containing demographic, income, crime, and survey-response information relevant to that

location. (Figure 7) If a county has no respondents, or if the survey respondent layer is turned off, the pop-up shows only county-level characteristics. This structure enables efficient exploration of spatial differences and helps users identify how social, economic, and population factors align geographically.

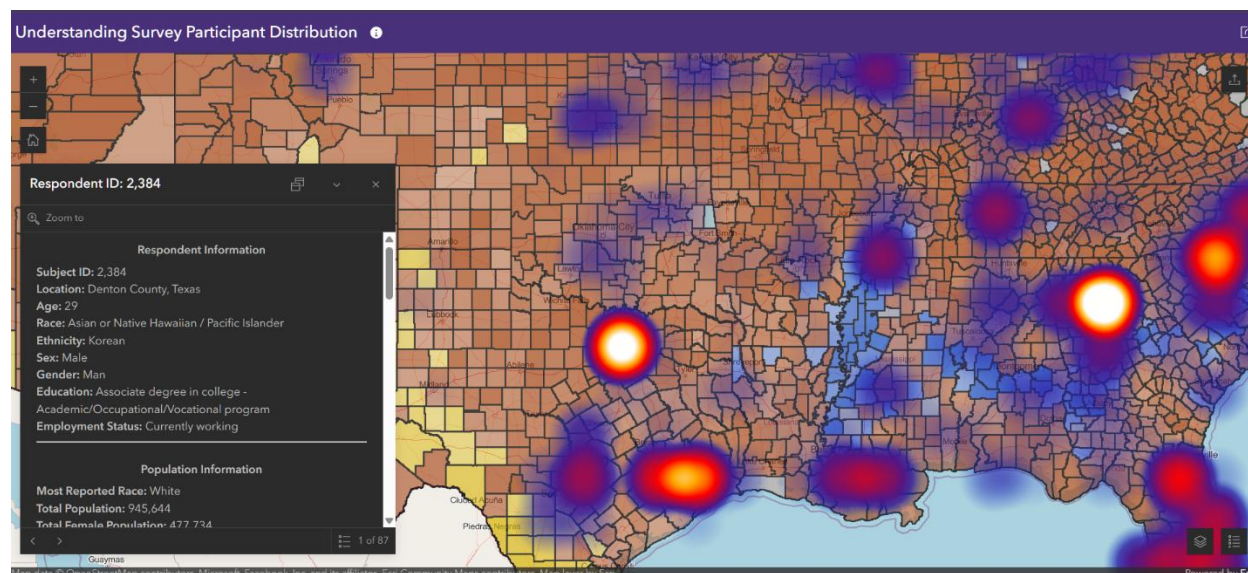


Figure 7: Interactive ArcGIS Pop-Up Display

3.2 R Shiny Application

The Survey Explorer application allows users to explore survey and county-level data through interactive tables and visualizations. It consists of a landing page and three additional tabs according to Figure 6: *Visualizations*, *All Counties*, and *Individual County*.

Figure 8 is the instruction page that introduces the layout and explains how to navigate the application. It outlines the purpose of each tab and provides basic guidance such as hovering over plots to view exact percentages and noting that margins of error exist in the raw data files.

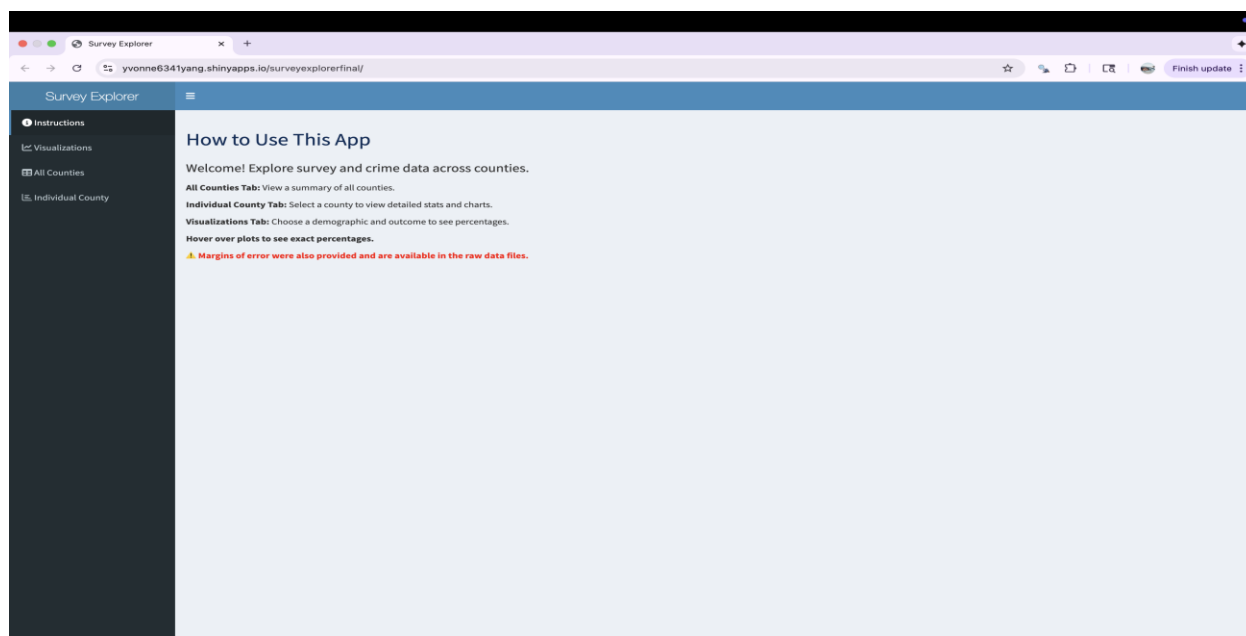


Figure 8: RShiny Landing/Instruction Page

The *Visualization* tab allows users to explore how mental-health outcomes vary across demographic groups (Figure 9). Two dropdown menus allow users to select a demographic variable (e.g., sex, age, race, education) and an outcome variable (e.g., depression/anxiety, self-hard, or suicidal ideation within 12 months) (Figure 10). Once both are selected, the application automatically generates an interactive bar chart displaying the percentage of each subgroup that falls into the chosen categories. Hovering over each bar reveals the exact percentage, allowing users to compare distributions quickly and intuitively across demographic groups.

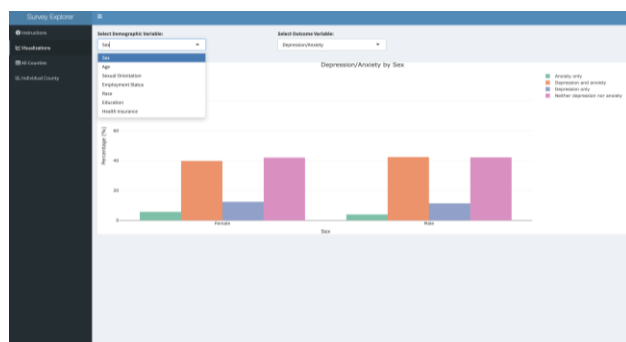


Figure 9: Rshiny Visualizations Tab – Demographics

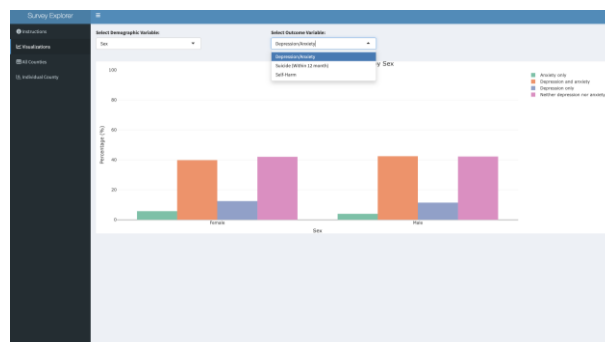


Figure 10: Rshiny Visualizations Tab – Outcome Variables

The *All Counties* tab provides an interactive summary table for all counties represented in the dataset (Figure 11). Users can search by state, filter variables, and sort columns in ascending or descending order. The table integrates demographic, population, income, and crime indicators, helping users identify trends or outliers across counties.

Survey Explorer

Instructions
Visualizations
All Counties
Individual County

All Counties Summary

Texas

County	Participants	Total Population ↓	Black Population	Asian Population	Two or More Population	Median Income
Denton County, Texas	82	945,644	101,626	97,666	124,311	\$108,185
Hidalgo County, Texas	3	880,921	6,070	8,981	419,107	\$52,281
El Paso County, Texas	5	866,275	28,593	10,988	336,573	\$58,859
Galveston County, Texas	2	354,721	43,820	11,842	53,396	\$85,348
Hays County, Texas	21	256,429	10,507	4,947	52,273	\$85,827
Jefferson County, Texas	5	253,939	84,147	9,552	30,226	\$59,934
Midland County, Texas	1	171,496	12,427	3,715	29,611	\$93,984
Randall County, Texas	1	143,791	5,230	2,796	17,175	\$80,905
Hunt County, Texas	2	104,917	8,392	1,123	9,918	\$70,112
Liberty County, Texas	49	97,993	7,907	522	13,034	\$64,773
Waller County, Texas	33	59,455	14,046	996	7,213	\$76,135
Anderson County, Texas	1	57,825	10,860	330	9,359	\$58,846

1–12 of 23 rows

Previous 1 2 Next

Note: If the median income for a specific race is 0, it means there are no households in that county with a major household owner identified as that race.

Figure 11: Rshiny All Counties Tab

The *Individual County* tab allows users to select a county from a dropdown menu and view a detailed summary for that location (Figure 12). The top of the screen presents key county statistics, including population counts, racial breakdowns, median incomes, and major crime categories. Three bar charts display the following information:

- The racial composition of the county as a percentage of total population
- Median income percentages by racial group
- Relative percentage of different crime types

The users can download plots for inclusion in reports or presentations.

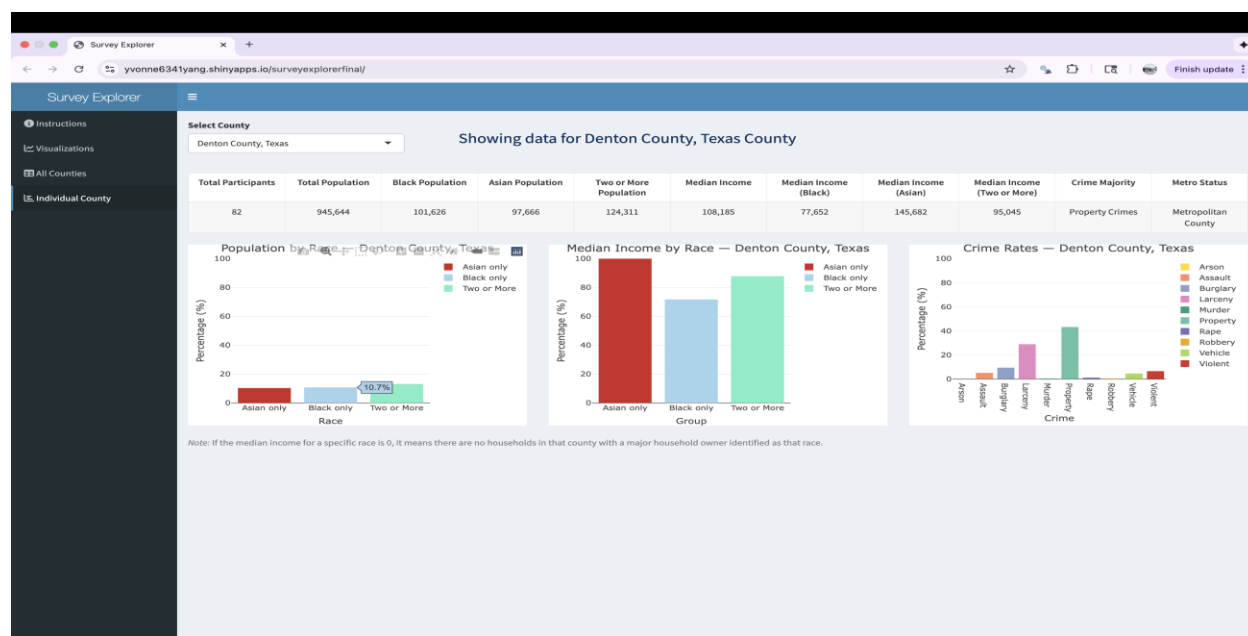


Figure 12: RShiny Individual County Tab

4. Discussion

ArcGIS Pro and R Shiny function together to display and interpret the integrated dataset. Both tools translate county-level demographic, income, crime, and mental-health information into accessible visual formats that support interactive exploration and interpretation. The ArcGIS map focused geographical patterns, while the R Shiny application clarified how those same patterns vary across demographic and socioeconomic groups.

The county-level maps illustrated consistent patterns across the United States. Crime rates were highest in urban and densely populated areas, whereas large regions of the Midwest and Great Plains displayed lower reported crime. These findings provide a contextual foundation for interpreting social environments relevant to mental health. The majority-race map simplified demographic complexity, showing White-majority populations across rural regions and Black or Asian majorities concentrated in urban or coastal areas. Median household income followed

clear regional divides: higher in the Northeast, West Coast, and suburbs, and lower in the South and rural Midwest. Survey responses clustered mainly in metropolitan areas, highlighting uneven coverage that should be considered when interpreting comparisons.

Building on these geographic insights, the R Shiny application provided an interactive view of how mental-health experiences differ across population groups. Gender patterns indicated similar prevalence of depression and anxiety but diverging severity: males reported higher rates of suicide attempts (12.6%) and self-harm (19%), while females more often experienced suicidal ideation only (12.4%). These results suggest that distress manifests differently by gender, underscoring the need for gender-sensitive mental health strategies.

Age trends revealed two stages when anxiety, depression, and self-harm were most frequent: early adulthood and the late twenties. Differences by sexual orientation were more pronounced. Respondents identifying as homosexual and bisexual reported higher levels of suicidal ideation than heterosexual respondents, with suicidal ideation reported by 19.7% of homosexual respondents and 22.2% of bisexual respondents, compared with 9.3% among heterosexual respondents. Similar patterns were observed for self-harm, reported by 18% of homosexual respondents and 25.4% of bisexual respondents, relative to 15.4% among heterosexual respondents. Employment status further differentiated mental-health outcomes. Respondents on disability leave and medical leave reported the highest combined depression and anxiety levels, at 53.5% and 58.1%, respectively. In contrast, respondents who were unemployed and not seeking work exhibited the highest rate of suicide attempts, at 20.5%. These distributions indicate that mental-health outcomes vary systematically across employment conditions, particularly among groups facing health-related or economic constraints.

Racial, educational, and insurance-based disparities added additional dimensions to this variation. Depression and anxiety were similar across races, but Black or African American respondents reported fewer suicide attempts (5.8%) and less self-harm (87.8%), while those identifying with two or more races exhibited the most suicidal ideation. Education level was associated with variation in mental-health outcomes. Respondents with doctoral degrees and professional degrees reported the highest rates of suicide attempts, at 44.4% and 34.2%, respectively. In contrast, respondents with less than a high school education showed the highest level of suicidal ideation, reported by 17.4% of this group. These patterns indicate that more severe outcomes differed across educational levels, with suicide attempts more prevalent among highly educated respondents and suicidal ideation more common among those with lower educational attainment. Insurance coverage also reflected structural inequality: individuals with both public and private insurance reported the highest depression and anxiety (50.5%), while those without insurance showed the most suicidal ideation (15.6%) but fewest attempts (4.4%), indicating potential barriers to accessing care.

At the county level, participation varied widely. Bergen County, New Jersey, contained the largest number of respondents (251 participants), characterized by high median income (\$123,715), a majority Asian population (159,137), and arson as its prevalent crime category. However, over one-third of counties had only one participant, and nearly two-thirds had fewer than five, limiting generalizability. Expanding sample coverage would improve representativeness and analytical reliability in future studies.

5. Conclusion

This project developed an integrated visualization framework that combined ArcGIS Pro mapping with an R Shiny dashboard to support descriptive exploration of mental-health survey data within county-level social and economic contexts. The ArcGIS maps provided spatial context for understanding where differences occur across counties, while the R Shiny application enabled interactive examination of demographic and socioeconomic variation in mental-health outcomes. Although uneven survey participation across counties limited generalizability, the visualizations revealed consistent patterns across sexual orientation, employment status, and education level. Beyond summarizing results, the Shiny dashboard supported flexible exploratory data analysis by allowing users to filter, compare, and examine subgroup patterns dynamically. Future work could strengthen this framework by expanding geographic coverage, incorporating additional contextual data sources, and applying longitudinal or predictive methods to examine how environmental influences on mental health evolve over time.

Appendix A: Variable Recoding and Category Consolidation Rules

Table A1. *Recoding of Race Variable for Visualization*

Original Response Pattern	Final Race Category	Recoding Rule
Black or African American	Black	Respondent selected only Black or African American
Asian or Native Hawaiian / Pacific Islander	Asian	Respondent selected only Asian or Native Hawaiian / Pacific Islander
Any response containing Black or African American and one or more additional race or ethnicity categories	Two or More Races	Multiple racial or ethnic identities selected
Any response containing Asian or Native Hawaiian / Pacific Islander and one or more additional race or ethnicity categories	Two or More Races	Multiple racial or ethnic identities selected
Any response containing two or more distinct race or ethnicity categories (e.g., White, Hispanic or Latino, Middle Eastern or North African, American Indian or Alaska Native)	Two or More Races	Multiple selections regardless of specific combination

Table A2. *Recoding of Education Variable for Visualization*

Original Survey Response	Recoded Category	Recoding Rule
Less than a high school diploma	Less than High School	Retained as lowest education category
High school diploma or equivalent (GED)	High School	Collapsed to represent secondary education
Some college but no degree	High School	Combined with high school to reduce sparsity and reflect incomplete postsecondary education
Associate degree in college – Academic/Occupational/Vocational program	Associate	Retained as distinct postsecondary credential
Bachelor's degree (e.g., BA, BS)	Bachelor	Retained as distinct undergraduate degree
Master's degree (e.g., MA, MS, MSW, MEd, MBA)	Master	Retained as distinct graduate degree
Doctorate (e.g., PhD, EdD)	Doctorate	Retained as highest academic degree
Professional school degree (e.g., MD, DDS, DVM, JD)	Professional Degree	Retained as advanced professional training

Table A3. *Recoding of Health Insurance Variable for Visualization*

Original Survey Response	Recoded Category	Recoding Rule
Yes, I am covered by a government health plan (e.g., Medicare, Medicaid)	Government Insurance	Classified as public insurance coverage
Yes, I have private health insurance	Private Insurance	Classified as private insurance coverage
Yes, I have both private insurance and government coverage	Both Government and Private	Classified as dual coverage
No, I do not have health insurance	No Insurance	Classified as uninsured

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